Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r,\theta,t) = \frac{p_0 \cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$

$$\boldsymbol{A}(r,\theta,t) = -\frac{\mu_0 p_0 \omega}{4\pi r}\sin\left[\omega\left(t-\frac{r}{c}\right)\right]\hat{z}$$

satisfy the Lorenz gauge condition.

Solution:

The Lorenz gauge condition is

$$\boldsymbol{\nabla}\cdot\mathbf{A} = -\mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Calculating the time derivative of V

$$\frac{\partial V}{\partial t} = \frac{p_0 \cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$

Calculating the divergence of $\mathbf{A}$

$$\boldsymbol{\nabla}\cdot\mathbf{A} =$$

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Problem 11.2

Problem 11.3

Problem 11.4

Problem 11.5

Problem 11.6

Problem 11.7

Problem 11.8

Problem 11.9

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Problem 11.11

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Problem 11.35

Problem 11.36